

only a reduced, unitary matrix $\hat{S}$ with diagonal elements $\eta_i (= |S_{ii}|)$.

Ignoring the so-called trivial ambiguities in the solutions for the phases of the off-diagonal elements [6], [7], the symmetric matrix $\hat{S}$ can indeed be reconstructed uniquely from knowledge of $\{\eta_i\}$; it should be added here that this is the only case for which there is a one-to-one correspondence between the sets $\{\eta_i\}$ and $\{\chi_{ij}\}$ ($\chi_{ij} = |\hat{S}_{ij}| = |S_{ij}|$), and the diagonal theorem [i.e., the four inequalities in (13a) defining the region $\mathcal{D}_1(\eta)$] provides a complete description of the unitary matrix.

In the nonsymmetric case, one has additional degrees of freedom and the phases of two of the off-diagonal elements of $\hat{S}$ can be chosen freely (with certain restrictions on the pairing); however, given the two phases and the values of η_i , all possible forms of $\hat{S}$ can in principle be constructed. This follows from the fact that $\{s_i\}$, and thus the bounds on $|d|$ stated in (19) and (20), can be calculated from $\{\eta_i\}$; (9) shows that the magnitudes of the off-diagonal elements can now be varied continuously within certain limits determined by the actual values of η_i .

For $n \geq 4$, the constraints on $\{\chi_{ij}\}$ cannot be derived from the diagonal theorems [2]–[4]; to substantiate this, we consider the general symmetric case, which ought to be the simplest one: The nonlinear equations can now be written as ($i, j = 1, 2, \dots, n$)

$$\eta_i^2 + \sum_{\substack{k=1 \\ k \neq i}}^n \chi_{ki}^2 = 1 \quad (22a)$$

$$\eta_i e^{i\alpha_{ij}} - \eta_j e^{-i\alpha_{ij}} = i \sum_{\substack{k=1 \\ (k \neq i, j; i < j)}}^n \left(\frac{\chi_{ki} \chi_{kj}}{\chi_{ij}} \right) e^{i(\alpha_{kj} - \alpha_{ki})}. \quad (22b)$$

To specify the matrix $\hat{S}$, we need $\frac{1}{2}n(n-1)$ real, independent

parameters, and it is readily seen that the set $\{\chi_{ij}\}$ contains just the right number of quantities. Since the diagonal elements can be calculated directly from (22a), the problem is therefore to determine the range of $\{\chi_{ij}\}$, subject to the restrictions imposed by (22a), for which the system of equations (22b) has solutions for $\{\alpha_{ij}\}$. Evidently, the dimension of the set $\{\chi_{ij}\}$ is larger than that of $\{\eta_i\}$ for $n \geq 4$, so the known constraints on $\{\eta_i\}$ will not yield the sufficient conditions on $\{\chi_{ij}\}$ for such values of n ; indeed, no complete solution of this problem has so far been given [7].

Note added in proof:

After the completion of this work, a new algorithm for parametrization of unitary ($n \times n$) matrices has been published by P. Diŕă: "Parametrisation of unitary matrices," *J. Phys. A: Math. Gen.*, vol. 15, pp. 3465–3473, 1982.

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Circuits and Systems Letters

Self-Calibration Technique for A/D Converters

Abstract—Linearity errors of weighted capacitor digital-to-analog converters (DAC's) can be corrected using a simple digital algorithm. The additional circuitry required is approximately 350 digital gates plus 10 bytes of memory space.

I. INTRODUCTION

Digital audio systems and certain data acquisition systems require analog-to-digital (A/D) converters with resolution and integral linearity corresponding to 14 bits or more and the sampling rates on the order of 50 kilosamples/s. However, converter resolution and integral linearity above 12 bits have been difficult to achieve in fast monolithic integrated circuit implementations. Laser-trimmed resistor networks are costly in terms of fabrication processing and chip area, and suffer from drift with time.

Previous self-calibration techniques require a separate calibration reference source and sometimes a large memory space to store nonlinearity information [1], [2]. Here we describe a self-calibration technique which requires no laser trimming, no calibration reference digital-to-analog converter (DAC), and only 10 bytes of memory. The technique has been proposed independently by Domogalla [3].

II. SELF-CALIBRATION TECHNIQUE

A block diagram of the new self-calibrating successive-approximation A/D converter is shown in Fig. 1. A metal-oxide-semiconductor (MOS) weighted-capacitor DAC, which inherently has relatively good integral linearity, is employed as DAC for the most significant bits (*main DAC*) [4], [5]. The less significant bits are provided by a resistor-string *sub DAC*, which is inherently free of differential nonlinearity [6]. An initial calibration cycle, typically performed each time power is turned on, corrects the linearity error of the main DAC through use of the *calibration DAC*.

Digital control circuits govern capacitor switching during the calibration cycle, and store the nonlinearity correction terms in data registers. The ratio errors of the sub DAC, and overall quantization errors accumulate during digital computation of error voltages. To overcome these errors for a 16-bit converter, 2

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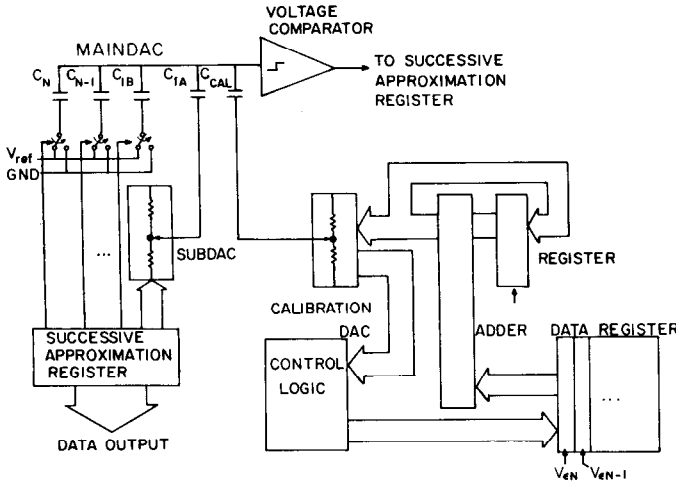


Fig. 1. Block diagram of self-calibrating A/D converter.

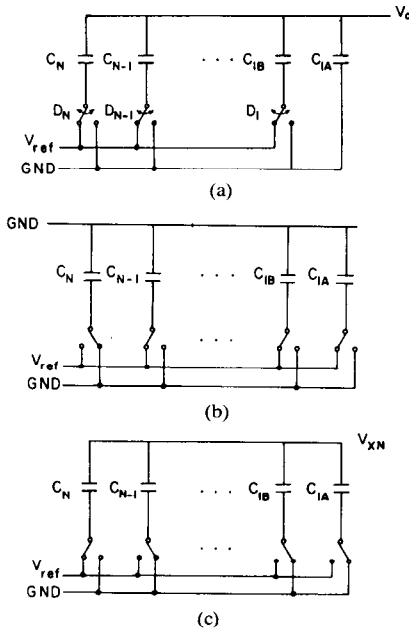


Fig. 2. (a) Weighted capacitor DAC. (b) Precharge cycle. (c) Charge redistribution and error acquisition.

bits of additional resolution is needed during the calibration cycle.

III. DETAILS OF OPERATION

Fig. 2(a) shows an N -bit weighted capacitor DAC. Each weighted capacitor C_n is assumed to be off by a factor of $(1 + \epsilon_n)$ from the ideal value due to process variations:

$$C_n = 2^{n-1}C(1 + \epsilon_n), \quad n = 1A, 1B, \dots, N \quad (1)$$

where the unit capacitor C is defined as the total capacitance divided by 2^N

$$C = \frac{C_{\text{total}}}{2^N} = \frac{C}{2^N} \sum_{i=1A}^N 2^{i-1}(1 + \epsilon_i) = C + \sum_{i=1A}^N 2^{i-1}\epsilon_i \quad (2)$$

and we get

$$\sum_{i=1A}^N 2^{i-1}\epsilon_i = 0. \quad (3)$$

The output voltage of a weighted capacitor DAC such as shown in Fig. 2(a) can be expressed in terms of capacitor values

and corresponding digital input codes:

$$V_o = \frac{V_{\text{ref}}}{C_{\text{total}}} \sum_{i=1B}^N C_i D_i = \frac{V_{\text{ref}}}{2^N \left(1 + \sum_{i=1A}^N 2^{i-1}\epsilon_i \right)} \sum_{i=1B}^N 2^{i-1}(1 + \epsilon_i) D_i$$

and by (3),

$$V_o = \frac{V_{\text{ref}}}{2^N} \sum_{i=1B}^N 2^{i-1}(1 + \epsilon_i) D_i. \quad (4)$$

The ideal output voltage $V_{o,\text{ideal}}$ can be obtained by taking $\epsilon_i = 0$ for all i in (4)

$$V_{o,\text{ideal}} = \frac{V_{\text{ref}}}{2^N} \sum_{i=1B}^N 2^{i-1} D_i. \quad (5)$$

The error voltage V_{error} is the difference between ideal and actual output voltages:

$$V_{\text{error}} = V_o - V_{o,\text{ideal}} = \frac{V_{\text{ref}}}{2^N} \sum_{i=1B}^N 2^{i-1}\epsilon_i D_i. \quad (6)$$

Define the error voltage $V_{\epsilon n}$ due to the n th capacitor mismatch

$$V_{\epsilon n} \equiv \frac{V_{\text{ref}}}{2^N} 2^{n-1}\epsilon_n, \quad n = 1B, 2, \dots, N. \quad (7)$$

Then, (6) becomes

$$V_{\text{error}} = \sum_{i=1B}^N V_{\epsilon i} D_i. \quad (8)$$

The calibration cycle begins by measuring the nonlinearity due to the MSB capacitor C_N . This is done by sampling the reference voltage V_{ref} on all the capacitors except the MSB capacitor, as shown in Fig. 2(b). At this time, the sampled charge in the capacitor array is

$$Q = -V_{\text{ref}} \sum_{i=1A}^{N-1} C_i = -CV_{\text{ref}} \sum_{i=1A}^{N-1} 2^{i-1}(1 + \epsilon_i) \quad (9)$$

Next, charge is redistributed by reversing the switching configuration, as shown in Fig. 2(c). The residual charge at the top plate of the capacitor array is

$$Q_{xN} = V_{\text{ref}} \left(C_N - \sum_{i=1A}^{N-1} C_i \right) = CV_{\text{ref}} \left(2^{N-1}\epsilon_N - \sum_{i=1A}^{N-1} 2^{i-1}\epsilon_i \right). \quad (10)$$

From (3), it follows

$$\sum_{i=1A}^N 2^{i-1}\epsilon_i = 2^{N-1}\epsilon_N + \sum_{i=1A}^{N-1} 2^{i-1}\epsilon_i = 0.$$

Thus

$$\sum_{i=1A}^{N-1} 2^{i-1}\epsilon_i = -2^{N-1}\epsilon_N. \quad (11)$$

Substituting (11) in (10), we get

$$Q_{xN} = CV_{\text{ref}} 2^N \epsilon_N \quad (12)$$

The residual voltage V_{xN} is

$$V_{xN} = \frac{Q_{xN}}{C_{\text{total}}} = 2V_{\epsilon N} \quad (13)$$

Similarly, errors due to smaller capacitors are measured. In each case, a successive approximation search using the sub DAC is used.

It can be easily shown that the general relation between residual voltages (V_{xn} 's) and the error voltages (V_{ϵ} 's) is

$$V_{\epsilon n} = \frac{1}{2} \left(V_{xn} - \sum_{i=n+1}^N V_{\epsilon i} \right), \quad n = 1, 2, \dots, N-1 \quad (14)$$

or in digital domain,

$$DV_{\epsilon N} = \frac{DV_{xN}}{2} \quad (15)$$

$$DV_{\epsilon n} = \frac{1}{2} \left(DV_{xn} - \sum_{i=n+1}^N DV_{\epsilon i} \right), \quad n = 1, 2, \dots, N-1 \quad (16)$$

where $DV_{\epsilon n}$ and DV_{xn} stand for digitized error voltages (*error terms*) and digitized residual voltages (*residual terms*), respectively.

Therefore, by digitizing residual voltages using the sub DAC, error terms $DV_{\epsilon N}, DV_{\epsilon N-1}, \dots, DV_{\epsilon 1B}$ can be computed subsequently by (15) and (16). All these error terms are stored in digital memory.

During subsequent normal conversion cycles, the calibration logic is disengaged. The converter works the same way as an ordinary successive approximation converter, except that error-correction voltages are added by proper adjustment of the sub DAC digital input code. When the n th bit is being tested, corresponding error term $DV_{\epsilon n}$ is added to the error terms accumulated from the first bit (MSB) through the $(n-1)$ th bit. If the bit decision is 1, the added result is stored in the accumulator. Otherwise, $DV_{\epsilon n}$ is dropped, leaving the accumulator with the previous result. The content of the accumulator is converted to an analog voltage by the calibration DAC. This voltage is then subtracted from the main DAC output voltage through the capacitor C_{cal} . The overall operation precisely cancels the nonlinearity due to capacitor mismatches by subtracting V_{error} in (8) from the actual output voltage V_o in (4). The only extra operation involved in a normal conversion cycle is one 2's complement addition per bit.

IV. CONCLUSIONS

A new technique for self-calibration of successive-approximation analog-to-digital (A/D) converters is proposed.

The combination of weighted-capacitor and resistor-string provides a DAC of high resolution without having too large ratios between the components. The nonlinearity of weighted capacitor DAC can be corrected to desired level of linearity. The calibration algorithm is simple enough not to add too much complexity to the original converter.

Analysis and simulation of a possible integrated circuit implementation suggest that a complete converter, built in complementary MOS technology, should be linear to $\frac{1}{2}$ LSB at 14–16 bits of resolution while being operated at 50 kilosamples/s.

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Highly Selective Three-Dimensional Recursive Beam Filters Using Intersecting Resonant Planes

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Abstract—It is shown that three-dimensional (3-D) recursive filters, having passbands contained within uniform-width slices in $\omega_1 - \omega_2 - \omega_3$ space may be used to realize highly selective low-order 3-D recursive beam filters. These filters are capable of selectively transmitting spatially planar 3-D signals in quadrants where at least one of the frequency components is negative in $\omega_1 - \omega_2 - \omega_3$ space.

In the field of image processing, it is convenient to describe a time-sequence of images in terms of a scalar signal $s(x, y, t)$, where x and y are spatial variables and t is the temporal variable. There are many potential applications, especially in seismic and biomedical processing, where it is desirable to enhance (that is, filter) such a signal. A common circumstance is the processing of a 3-D planar, or nearly planar, signal. We consider a signal $s(x, y, t)$ to be spatially planar if its 3-D Fourier transform

$$S(\omega_1, \omega_2, \omega_3) \equiv \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} s(x, y, t) \cdot \exp[-j(\omega_1 x + \omega_2 y + \omega_3 t)] dx dy dt$$

lies totally in a single 3-D line passing through the origin in $\omega_1 - \omega_2 - \omega_3$ space and nearly planar if this is nearly so. Clearly, reflected seismic, sonar, and radar waves are approximately planar signals. The question, therefore, arises as to how one might design linear shift-invariant recursive 3-D filters to selectively transmit spatially planar signals having particular directions of propagation.

Consider the 3-D Laplace transform input-output transfer function $H(s_1, s_2, s_3)$, having a corresponding steady-state sinusoidal frequency response gain function given by $M(\omega_1, \omega_2, \omega_3) \equiv |H(j\omega_1, j\omega_2, j\omega_3)|$. The requirement that a 3-D filter only transmit signals that have a particular direction of propagation implies that the passband of the filter surrounds a straight line passing through the origin in $\omega_1 - \omega_2 - \omega_3$ space. A cone filter and a beam filter, as shown in Fig. 1(a), have this property that their passbands surround such a straight line.

We propose here an algebraically straightforward method for obtaining 3-D beam filters having high gain selectivity. The basic idea is to realize the required beam-shaped 3-D passband response outside quadrants where $\omega_1, \omega_2, \omega_3$ have the same sign by cascading two 3-D recursive filters, T'_1 and T'_2 . Each cascaded filter has a passband that surrounds a plane in $\omega_1 - \omega_2 - \omega_3$. Consider two prototype 1-D transform transfer functions $T_1(s)$

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